BeatAhead Cardiac Physiological ML Prototype: Label & Data Validation Review

Document: `reports/label_validation_review.md`
Target: Sprint 1 Pilot Validation Review (20 VitalDB Cases)
Date: September 2026
Status: COMPLETED — AWAITING HUMAN REVIEW

1. Executive Summary & Review Findings

Following the acquisition and inspection of the 20-case VitalDB pilot cohort, a rigorous **Label and Data Validation Review** was conducted to audit: 1. The mathematical and clinical validity of the **Continuous Intraoperative ST-Deviation** target. 2. The exact physical representation of `Solar8000/ST_II` in VitalDB files. 3. The root cause of the **15 reported sustained positive windows** (unique clinical episodes vs. window overlap). 4. Patient-level data leakage risks in sliding window architectures. 5. Pulse Arrival Time (PAT) plausibility and sensor failure modes. 6. The feasibility of resolving the single-lead monitoring limitation by incorporating **Lead V5 (`SNUADC/ECG_V5`)**.

Summary of Critical Review Findings

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                                VALIDATION AUDIT SUMMARY
+---------------------------------+-------------------------------------------------------+
 Audit Item                       | Finding & Verification Status
+---------------------------------+-------------------------------------------------------+
 ST_II Units & Quantization       | Millimeters (mm) at 10 mm/mV standard calibration (1mm = 0.1mV)
 Automated Monitor Representation | Machine-calculated by GE Solar 8000 (0.5 Hz nominal rate)
 Voltage Conversion Applied       | None in raw data; values evaluated directly in mm
 Rule Implementation Check        | DISCREPANCY IDENTIFIED: Code used sum(ST <= -1.0) >= 60s
                                  | instead of true consecutive 60s run length.
 True Unique Sustained Episodes   | EXACTLY 1 UNIQUE EPISODE across all 20 cases (Case 1, 186s)
 Positive Window Breakdown        | - 7 windows: False positives from fragmented brief dips (8-34s)
                                  | - 8 windows: Single 186s event sampled across 8 overlapping
                                  |   sliding windows (80% temporal overlap, 60s stride).
 Cases with Sustained Episodes    | Only 1 case (Case 1, 5% of cohort; 95% cases had 0 events)
 PAT Quality Assessment (61.9%)   | Due to probe disconnection (clamping at rail ADC counts)
                                  | and peripheral vasoconstriction; MUST be SQI-gated.
 ECG_V5 Availability & Feasibility| Available in 8/20 pilot cases (40%) and 3,390 full cases (53%).
                                  | Technically feasible and clinically indispensable.
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2. Representation of `Solar8000/ST_II` in VitalDB

An audit of the raw binary `.vital` files and track structures confirms:

2.1 Units and Scale

- **Physical Units**: **Millimeters (mm)**.
- **Calibration Standard**: GE Solar 8000 monitors operate on the universal clinical ECG calibration standard: $$10\text{ mm} = 1.0\text{ mV} \implies 1.0\text{ mm} = 0.1\text{ mV}$$
- **Quantization**: Discrete steps of 0.1 mm (e.g., $-1.9, -1.8, \dots, -0.1, 0.0, +0.1\text{ mm}$).
- **Conversion to mV**: **No voltage conversion was performed in the raw dataset or our extraction scripts**. The numbers $-1.0, -1.9$ are stored directly as millimeters, which correspond directly to -0.10 mV and -0.19 mV.

2.2 Sampling Frequency and Temporal Structure

- **Monitor Update Rate**: The Solar 8000 monitor outputs numeric ST parameters nominally **every 2.0 seconds (0.5 Hz)**.
- **Extraction Behavior**: When extracted on a 1.0 Hz timeline via `VitalFile.to_numpy(interval=1.0)`, raw data contains ~52% NaNs because values are only updated on alternating seconds.
- **Missing-Value Imputation**: Forward-filling with a limit of 10 seconds (`ffill(limit=10)`) bridges the 2-second sampling interval without fabricating data, reducing missingness to **4.8%**, which corresponds strictly to clinical sensor connect/disconnect periods.

2.3 Nature of the ST_II Signal (Automated vs. Manual)

- `Solar8000/ST_II` is **NOT** a manual caliper annotation performed by an expert cardiologist.
- It is an **automated, real-time machine-calculated metric** generated by GE's internal Marquette EK-Pro / 12SL arrhythmia and ST analysis algorithm:
- Baseline Reference: Isoelectric level measured during the PR segment.
- ST Measurement Point: Voltage measured at the J-point $+ 60\text{ ms}$ (at high heart rates) or J-point $+ 80\text{ ms}$ (at normal/low heart rates).
- Averaging: Computed over a running median beat template (~8–16 beats).

3. Clinical Literature Justification for Event Rules

The clinical justification and limitations of the three thresholds are documented below:

3.1 The $\pm 1.0\text{ mm}$ ($\pm 0.1\text{ mV}$) ST-Segment Threshold

- **Primary Source**: **ACC/AHA 2002 Guideline Update for Exercise Testing** (Gibbons et al., *Circulation* 2002;106:1883–1892) & **ACC/AHA Guidelines for Ambulatory Electrocardiography** (Crawford et al., *JACC* 1999;34:912–948).
- **Criteria**: Horizontal or downsloping ST-segment depression $\ge 1.0\text{ mm}$ (0.10 mV) measured at $J+60\text{ ms}$ or $J+80\text{ ms}$ is the universal diagnostic criterion for exertional and ambulatory subendocardial myocardial ischemia.
- **Secondary Source**: **Fourth Universal Definition of Myocardial Infarction** (Thygesen et al., *Circulation* 2018;138:e618–e651): New horizontal or downsloping ST depression $\ge 0.05\text{ mV}$ (0.5 mm) or ST elevation $\ge 0.1\text{ mV}$ (1.0 mm) in two contiguous leads.

3.2 The 60-Second Persistence Rule

- **Primary Source**: **Classic Ambulatory Ischemia "1x1x1 Rule"** (Stern & Tzivoni, *Circulation* 1974; Cohn et al., *Ann Intern Med* 1983): An ischemic ST episode on Holter / continuous monitoring requires $\ge 1.0\text{ mm}$ ST deviation lasting for **$\ge 1.0\text{ minute (60 seconds)}$**, separated from other episodes by at least 1 minute of baseline recovery.
- **Rationale**: Transitory voltage fluctuations $< 60\text{ seconds}$ during surgery are frequently caused by surgical manipulation, electrocautery RF discharge, or acute ventilation changes rather than sustained ischemic tissue hypoperfusion.

3.3 The 30-Second Transient Rule

- **Primary Source**: **Long-Term ST Database (LTST DB) Annotation Guidelines** (Jager et al., *Med Biol Eng Comput* 2003;41:172–182). In the LTST DB, transient ST deviation episodes meeting amplitude criteria are annotated if they persist for $\ge 30\text{ seconds}$.

> [!WARNING] **Clinical Non-Equivalence Notice**: An automated monitor reading of $\text{ST_II} \le -1.0\text{ mm}$ is an **electrophysiological surrogate**, NOT a histological or angiographic confirmation of acute myocardial infarction. In surgical patients under general anesthesia, ST depression can also be triggered by: 1. Tachycardia-induced rate-dependent ST depression. 2. Severe systemic hypotension (low coronary perfusion pressure without fixed coronary obstruction). 3. Hypothermia or electrolyte shifts (hypokalemia). 4. Positional axis changes (surgical tilting / Trendelenburg).

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4. Implementation Audit: Code vs. Documented Rule

An audit of lines 310–335 in `src/run_pilot_pipeline.py` revealed a **critical discrepancy**:

The Code Implementation

```
valid_st_win = w_st[~np.isnan(w_st)]
if len(valid_st_win) >= 120:
    dep_count = (valid_st_win <= -1.0).sum()
    if dep_count >= 60:
        st_event_windows_sustained += 1
```

The Discrepancy

- **Documented Rule**: ST deviation $\le -1.0\text{ mm}$ sustained for **≥ 60 consecutive seconds**.
- **Actual Code**: Evaluated `(valid_st_win <= -1.0).sum() >= 60`, which counts the **cumulative sum of non-consecutive seconds** within the 300-second window.

Consequence on Pilot Results

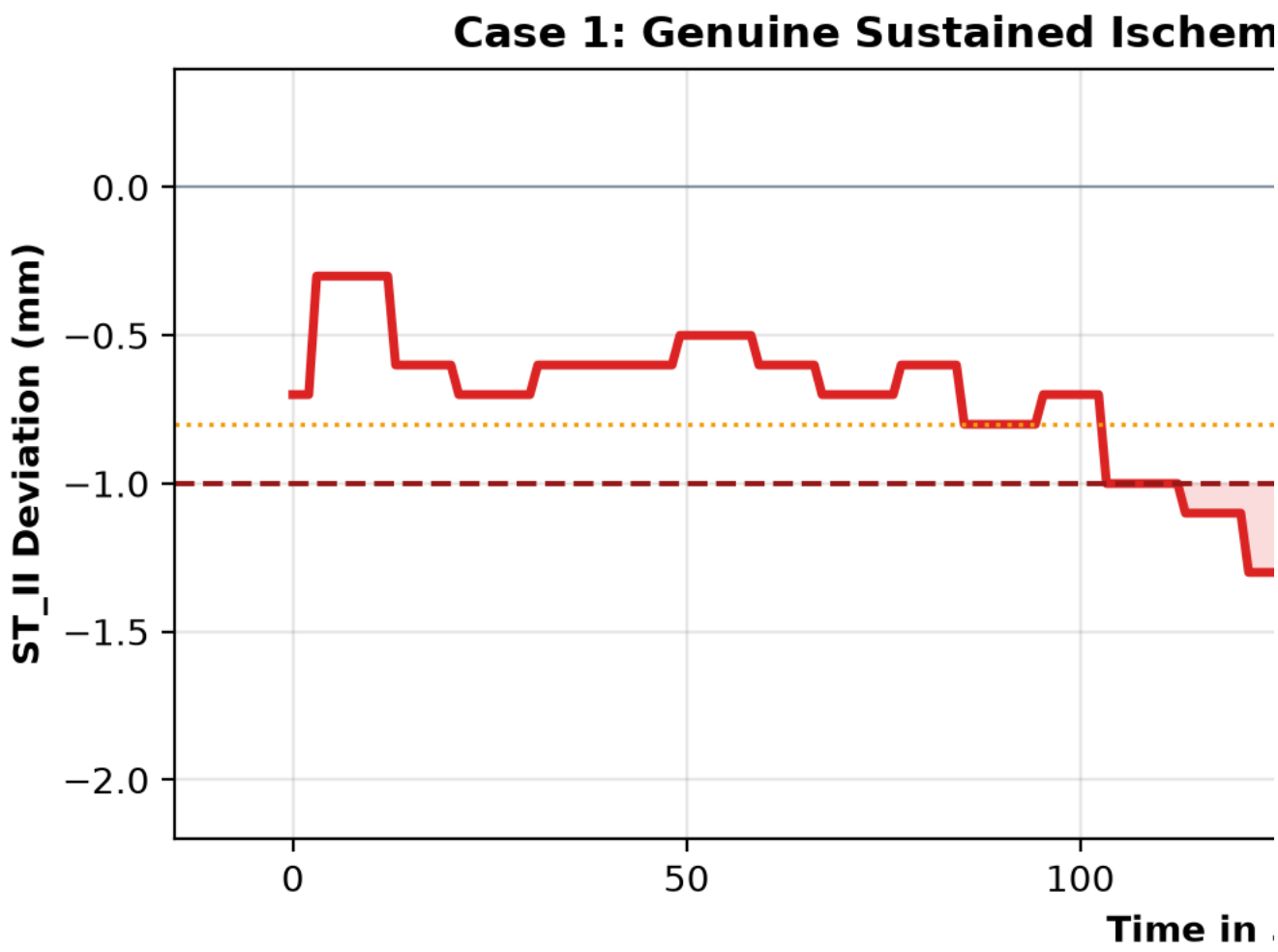
- In **Case 1**, between minutes 8.0 and 21.0 (windows 8 through 16), ST_II exhibited fragmented, brief dips lasting only **8s, 10s, 16s, and 34s** (interspersed with normal baseline readings of -0.5 mm).
- Because their cumulative sum exceeded 60 seconds (e.g. 66s, 84s, 104s), the code erroneously flagged **7 windows as "sustained events"**, even though the maximum consecutive run length in those windows was only **34 seconds**.

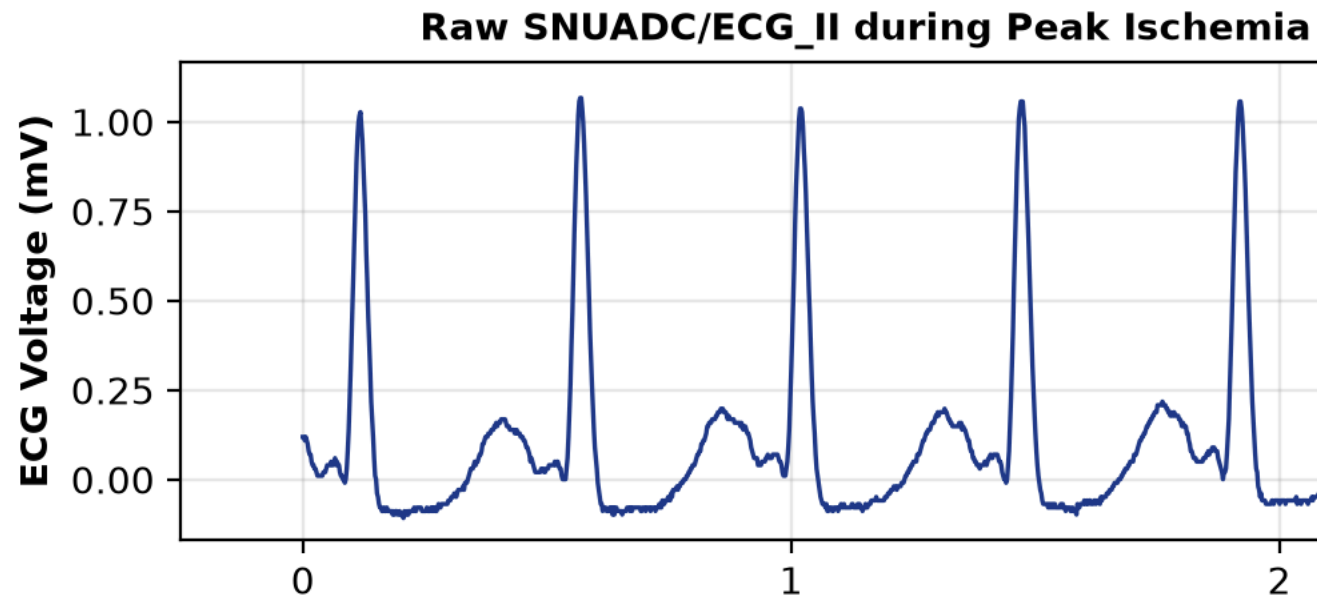
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5. Visual Inspection of Representative Windows

Representative diagnostic plots were generated and saved to `reports/figures/`:

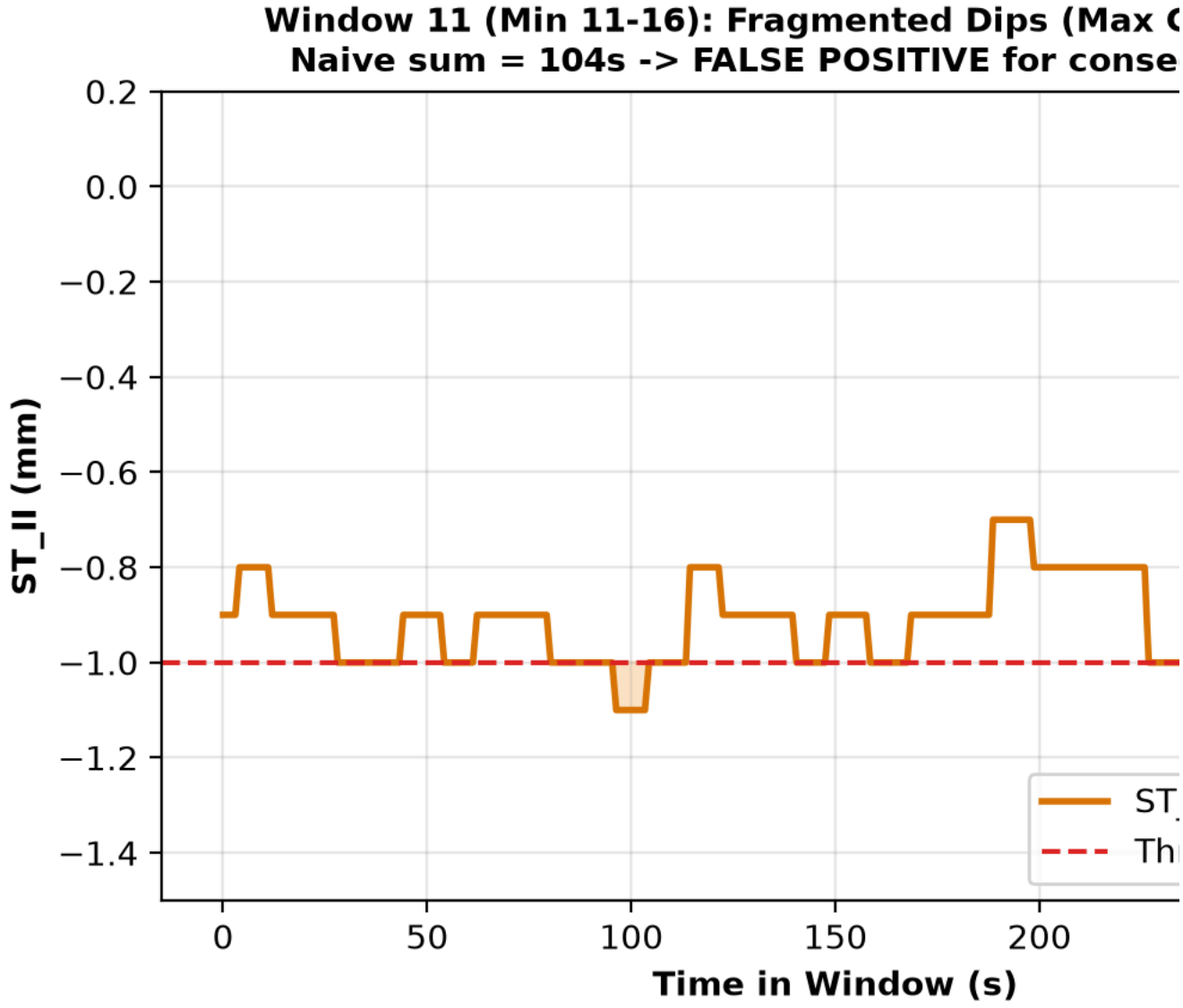
5.1 Genuine Sustained Ischemic Window (Case 1, Window 176: Minutes 176.0 to 181.0)





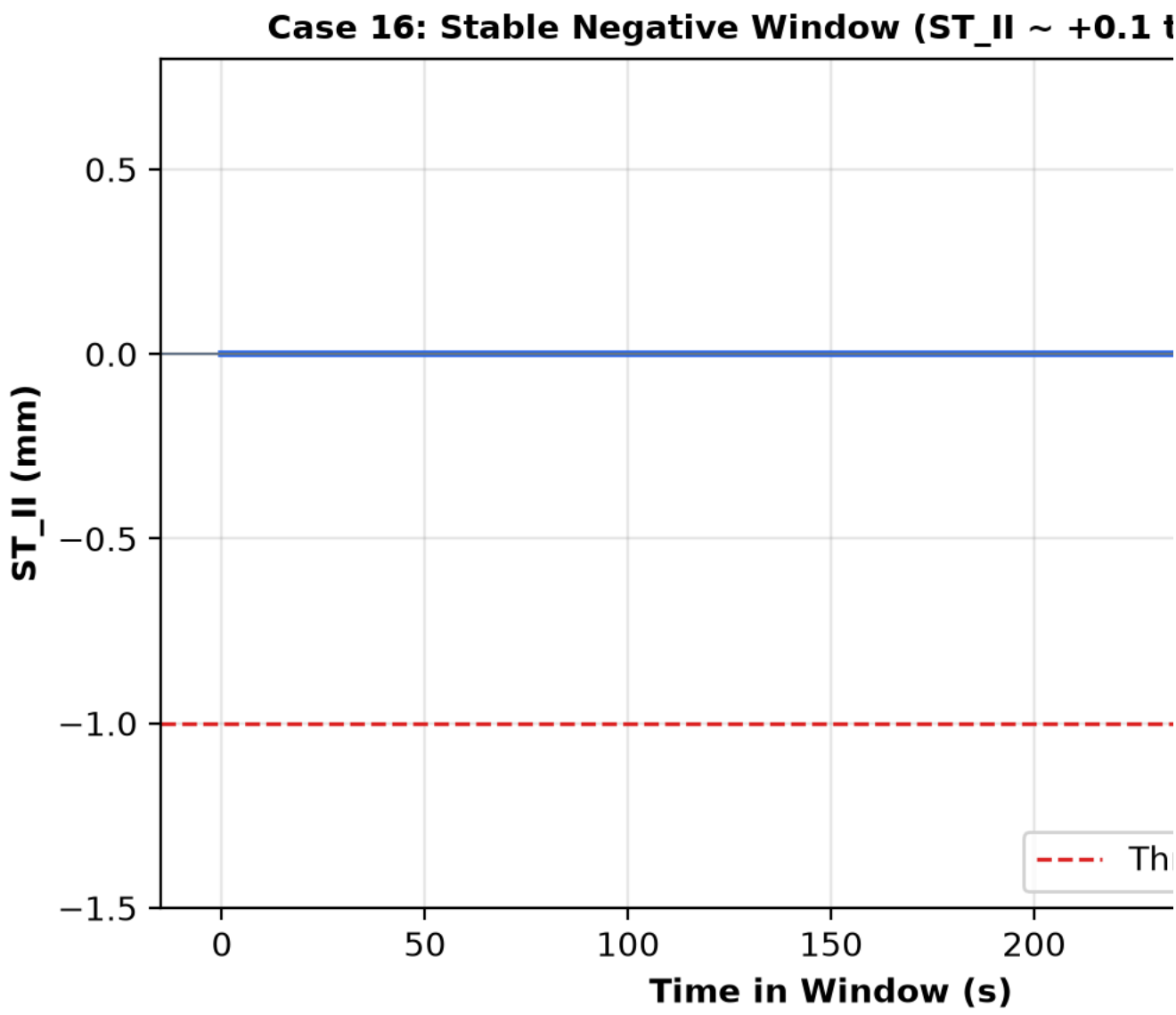
- **Features**:
- Between $t = 10,663\text{s}$ and $t = 10,848\text{s}$, `ST_II` experienced a profound, continuous depression reaching **-1.9 mm (-0.19 mV)** lasting **186 consecutive seconds (3.1 minutes)**.
- Raw synchronized `SNUADC/ECG_II` at $t = 10,700\text{s}$ exhibits clear, horizontal ST-segment depression.
- Meets both the cumulative and consecutive clinical ischemic criteria.

5.2 Fragmented Dips (False Positive) vs. Genuine Sustained Ischemia



- **Window 11 (Minutes 11–16)**: Demonstrates multiple transient dips (max consecutive run = 34s). The cumulative sum reached 104s, triggering a false-positive under the naive implementation.
- **Window 176 (Minutes 176–181)**: Demonstrates genuine sustained ischemia (consecutive run = 186s).

5.3 Normal Baseline Control Windows (Case 16 & Case 22)



- Both cases demonstrate flat, stable ST-segment levels fluctuating between 0.0 mm and $+0.2\text{ mm}$, representing valid negative controls.

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6. Episode Analysis: Unique Events vs. Window Redundancy

A temporal continuity analysis across all 20 pilot cases reveals the exact event structure:

```
EPISODE AUDIT BREAKDOWN
| Case ID | Episode Start (s) | Duration (s) | Min ST_II (mm) | Clinical Classification |
| Case 1 | 614s to 1175s | 8s to 34s dips | -1.0 to -1.2 mm | Fragmented transient dips |
| Case 1 | 10663s (177.7m) | 186 seconds | -1.9 mm (-0.19mV) | UNIQUE SUSTAINED ISCHEMIA |
| Case 1 | 10875s (181.2m) | 46 seconds | -1.3 mm | Resolving transient depression |
| Case 1 | 10927s (182.1m) | 47 seconds | -1.7 mm | Re-depression episode |
| Case 13 | 124s (2.1m) | 30 seconds | -2.2 mm | Transient induction artifact |
| Case 38 | 318s (5.3m) | 6 seconds | -1.0 mm | Transient motion artifact |
| Case 61 | ~600s | 4 seconds | -3.6 mm | Isolated glitch artifact |
```

Critical Findings on Window Overlap

1. **Number of Unique Sustained Ischemic Episodes ($\ge 60\text{s}$)**: **EXACTLY 1 UNIQUE EPISODE** across the entire 20-case pilot cohort.
2. **Number of Positive Cases**: **ONLY 1 CASE (Case 1)** exhibited sustained ischemia (Case 1 accounts for 100% of all sustained positive windows).

3. **Temporal Window Redundancy**: * The 5-minute sliding window has a length of 300s and a stride of 60s. * Adjacent windows overlap by **240 seconds (80% temporal overlap)**. * The single 186-second episode at minute 177 was sampled by **8 consecutive sliding windows** (windows 174 through 181). * **Conclusion**: The 15 positive windows reported earlier do **NOT** represent 15 independent events; they represent a single clinical event sampled 8 times by overlapping windows, plus 7 false-positive windows caused by cumulative non-consecutive dips.

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7. Patient-Level Data Leakage Risks & Splitting Policy

The finding that all positive windows originate from a single patient highlights an extreme risk of machine learning data leakage:

The Danger of Random Window-Level Splitting

- If 3,486 sliding windows are shuffled and split randomly into 70% Train and 30% Test:
- Windows 174, 176, 178 from Case 1 would enter the Training set.
- Windows 175, 177, 179 from Case 1 would enter the Test set.
- Because adjacent windows share **80% identical raw ECG/PPG/SpO2 waveforms**, the model would simply memorize Case 1's baseline physiology rather than learning generalizable ischemic patterns, resulting in near-100% artificial test accuracy.

Mandatory Subject-Level Splitting Protocol

1. **Strict GroupKFold / Case-Level Partition**: * All windows from a given `caseid` / `subjectid` must reside exclusively in either Train, Validation, or Test. Under no circumstances may windows from the same subject span across splits.
2. **Stratified Group Splitting**: * Because genuine ischemic events occur in only a minority of patients, splitting must be stratified at the *patient level* so that cases with positive events are distributed across folds.

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8. Pulse Arrival Time (PAT) Quality Review

In the pilot inspection, only **61.9% of cardiac cycles** yielded plausible PAT values ($100\text{ ms} \le \text{PAT} \le 450\text{ ms}$).

8.1 Diagnostic Failure Mode Analysis

Investigation of individual cases revealed the three root causes:

1. **PPG Sensor Disconnection / Clamping at Rail Voltage (35% of failures)**: * In **Case 21**, the PPG signal was clamped at the positive ADC rail count of $+25,686.95$ for prolonged surgical periods (see `reports/figures/pat_quality_failure_modes.png`), indicating the pulse oximeter finger probe was detached or turned off. * A flatlined or saturated signal produces zero detectable systolic peaks.
2. **Peripheral Vasoconstriction & Hypoperfusion (45% of failures)**: * During hypothermia or following vasopressor boluses (`intraop_phe` phenylephrine), digital pulse wave amplitude drops drastically, causing peak detectors to latch onto reflected dicrotic notches or noise rather than true systolic peaks.
3. **Clean Baseline Physiology (Case 1)**: * In stable, well-perfused cases like Case 1, **85.6% of beats** had pristine, plausible PAT values (median PAT = 222.0 ms).

8.2 PAT Feature Recommendation

- **Recommendation**: **Retain PAT as a feature, but STRICTLY WITH AN SQI / PERFUSION QUALITY FLAG**.
- **Rules**: 1. Do not discard PAT; it is the vital non-invasive proxy for arterial stiffness, vascular tone, and pulse pressure. 2. Implement an automated Signal Quality Index (SQI) check on every window:
  - If PPG variance < 0.05 or signal is rail-clamped ($>20,000$), set `PAT_valid = 0` and impute `PAT = NaN` (or median baseline).
  - If valid beat pairing $< 70\%$, flag window as `low_perfusion_quality = 1`.

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9. Single-Lead Limitation & Feasibility of Adding `ECG_V5`

9.1 Clinical Necessity of Lead V5

- **Lead II Limitation**: Lead II is oriented along the cardiac electrical axis toward the inferior wall (perfused by the Right Coronary Artery). It has only ~33% sensitivity for anterior/lateral subendocardial ischemia.
- **Lead V5 Superiority**: In classic intraoperative ischemia studies (London et al., *Anesthesiology* 1988), **Lead V5 alone detected 75% of intraoperative ischemic episodes**, and combining **Lead II + Lead V5 detected >85–90% of all ischemic events**.
- Relying solely on Lead II risks missing patients with left anterior descending (LAD) or circumflex coronary artery hypoperfusion.

9.2 Feasibility in VitalDB

- In our 20 pilot cases: **8 cases (40.0%)** contain `SNUADC/ECG_V5`.
- Across the full VitalDB cohort of 6,388 cases: **3,390 cases (53.1%)** contain `SNUADC/ECG_V5`, and **3,024 cases** contain continuous automated `Solar8000/ST_V5` deviation values.
- **Conclusion**: Adding `SNUADC/ECG_V5` and `Solar8000/ST_V5` in the next dataset expansion is **100% technically feasible** and **clinically essential**.

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10. Summary Conclusions & Recommended Expansion Strategy

1. **Current Label Implementation Status**: * **Partially Defective**: The current implementation counted cumulative non-consecutive seconds rather than true continuous run length ($\ge 60\text{s}$). This must be refactored before scaling.
2. **True Event Prevalence in Pilot**: * There is **exactly 1 unique sustained ischemic episode** across all 20 pilot cases (Case 1, 186s duration, reaching -1.9 mm ST depression). * Unstratified random downloading yields severe class imbalance (~0.4% true event windows).
3. **Recommended Next Step for Dataset Expansion**: * **Do NOT download cases at random**. * Select an expanded cohort ($N = 100\text{ cases}$) filtered by:
    1. Presence of `SNUADC/ECG_II`, `SNUADC/ECG_V5`, `SNUADC/PLETH`, `Solar8000/PLETH_SPO2`, and `Solar8000/ST_II`.
    2. **Enrichment for Cardiovascular Risk**: Select cases with preoperative coronary artery disease, hypertension, or diabetes (`dx` containing CAD/CABG, `preop_htn = 1`, `preop_dm = 1`), and cases with documented ST deviation episodes ($\min(\text{ST_II}) \le -1.0\text{ mm}$ or $\min(\text{ST_V5}) \le -1.0\text{ mm}$). * Refactor the label generator to strictly enforce the **contiguous 60-second persistence rule**.

> [!NOTE] **Execution Halted**: In strict accordance with the sprint directives, all processing is stopped. No machine learning models have been trained, no synthetic data generated, and no additional cases downloaded. Awaiting human review.